Name: Divyanka Sagolale
PRN: 24070521116
Sec: B

```
%{
#include <stdio.h>
int yylex(void);
int yyerror(const char *s);
%}

%token FOR ID NUM RELOP INC DEC

%%

stmt:
    FOR '(' init ';' cond ';' update ')' '{' '}'
    {
        printf("Valid FOR loop\n");
    }
    ;

init:
      ID '=' NUM
    | ID '=' ID
    ;

cond:
      ID RELOP NUM
    | ID RELOP ID
    ;

update:
      ID INC
    | ID DEC
    | INC ID
    | DEC ID
    ;

%%

int main(void)
{
    printf("Enter a FOR loop:\n");
    yyparse();
    return 0;
}

int yyerror(const char *s)
{
    printf("Invalid FOR loop\n");
    return 0;
}
```

Name: Divyanka Sagolale
PRN: 24070521116
Sec: B

```
%{
#include <stdio.h>
#include "y.tab.h"
%}

%%

"for"                       { return FOR; }
"++"                        { return INC; }
"--"                        { return DEC; }
"<="|">="|"=="|"!="|"<"|">" { return RELOP; }
[a-zA-Z_][a-zA-Z0-9_]*       { return ID; }
[0-9]+                      { return NUM; }
[ \t\n]+                    ;
.                           { return yytext[0]; }

%%

int yywrap(void)
{
    return 1;
}
```

Name: Divyanka Sagolale
PRN: 24070521116
Sec: B

```
div12@div12:~/CC lab$ yacc -d for.y
div12@div12:~/CC lab$ lex for.l
div12@div12:~/CC lab$ gcc y.tab.c lex.yy.c -o for
div12@div12:~/CC lab$ ./for
Enter a FOR loop:
for(i=0;i<10;i++) {}
Valid FOR loop
```